

CSCI 5244: Homework Set 3

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1. Given two vectors $x, y \in \mathbb{R}^n$, using amplitude encoding we can encode them as quantum states

$$|x\rangle = \sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|}$$

where $\|\cdot\|$ denotes the Euclidean norm and $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ are orthonormal quantum states.

- (a) Show that the inner product $x^T y = \|x\| \|y\| \langle x|y\rangle$

Begin solution

Given $\langle x|y\rangle$, we can define its amplitudes as the sum of the conjugate transpose of $|x\rangle$ which is written as $\langle x|$ and $|y\rangle$ which is $a_0^* b_0 + a_1^* b_1 + \dots + a_n^* b_n$

Rewrite the given amplitudes to reflect $\langle x|y\rangle$

$$\langle x| = \sum_{k=1}^n \frac{x_k}{\|x\|}, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|}$$

Now we can write $\langle x|y\rangle$ as the double sum

$$\langle x|y\rangle = \sum_{j=1}^n \frac{x_j}{\|x\|} \sum_{k=1}^n \frac{y_k}{\|y\|}$$

Since these are orthonormal quantum states for every pair (j, k) where $j \neq k$ results in a 0, else $j = k$ results in a 1. This is just the Kronecker Delta δ_{jk} , thus the sum reduces to

$$\langle x|y\rangle = \sum_i \frac{x_i y_i}{\|x\| \|y\|}$$

Thus we get

$$x^T y = \|x\| \|y\| \frac{1}{\|x\| \|y\|} \sum_{i=1}^n x_i y_i$$

$$x^T y = \sum_{i=1}^n x_i y_i$$

- (b) The outer product xy^T is an $n \times n$ matrix. How to encode xy^T quantumly using $|x\rangle$ and $|y\rangle$?

Begin solution

xy^T is an $n \times n$ matrix with (j, k) which can be represented quantumly as $|x\rangle \langle y|$, the outer product.

Expanding the outer product with given amplitudes gives us

$$|x\rangle = \sum_{j=1}^n \frac{x_j}{\|x\|}, \langle y| = \sum_{k=1}^n \frac{y_k}{\|y\|} \langle k|$$

Writing as a double sum

$$|x\rangle \langle y| = \frac{1}{\|x\|\|y\|} \sum_j \sum_k x_j y_k$$

This differs from $x_j y_k$ by the same constant $\frac{1}{\|x\|\|y\|}$ for every entry, so

$$A = \|x\|\|y\| |x\rangle \langle y|$$